MD HASANUR RAHMAN

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EDUCATION

University of Florida *2025–Present*
PhD in Electrical and Computer Engineering
Advisor: Guanpeng Li
Continued PhD after advisor's move from University of Iowa

University of Iowa *2021–2025*
PhD study in Computer Science
MS in Computer Science, 2024
Advisor: Guanpeng Li

Bangladesh University of Engineering and Technology *2015–2019*
BS in Computer Science and Engineering

RESEARCH INTERESTS

High Performance Computing, Dependable Computing Systems, Scientific Data Compression, Parallel Computing, Program Analysis, Large Language Model Inference

SELECTED RESEARCH CONTRIBUTIONS

- **Workload-Guided Protection for HPC Applications:** Developed techniques that analyze how different application parameters affect silent data corruption risk, enabling selective protection of vulnerable executions while maintaining very low overhead.
- **DNN Reliability under Hardware Faults:** Characterized how transient hardware faults affect deep neural network inference, distinguishing hardware-induced failures from algorithmic inaccuracy and developing scalable fault-injection support for modern DNN models.
- **Resource-Aware Scientific Data Compression:** Developed feature-driven and endpoint-aware frameworks that optimize how scientific datasets respond to lossy compression, helping balance data reduction, reconstruction quality, and downstream HPC application accuracy.
- **Reliable and Efficient AI Inference Systems:** Building on prior work in HPC resilience, scientific data compression, and GPU systems to develop large language model inference techniques that improve memory efficiency, execution reliability, and scalability for emerging AI workloads.

PUBLICATIONS

First-Author Peer-Reviewed Conference Publications

- **Deploying Lightweight Input-Aware Selective Instruction Duplication in HPC Applications**
  Md Hasanur Rahman, Guanpeng Li
  ACM International Conference for High-Performance Computing, Networking, Storage and Analysis (SC), 2025
  Acceptance rate: 21.2%
- **Modeling Rate-Distortion for Endpoint-Aware Lossy Compression in Scientific Data Transfer**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Performance Computing and Communications Conference (IPCCC), 2025
  Acceptance rate: 32.1%

- **DRUTO: Upper-Bounding Silent Data Corruption Vulnerability in GPU Applications**
  Md Hasanur Rahman, Sheng Di, Shengjian Guo, Xiaoyi Lu, Guanpeng Li, Franck Cappello
  IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2024
  Acceptance rate: 25.0%
- **A Generic and Efficient Framework for Estimating Lossy Compressibility of Scientific Data**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Conference on Massive Storage Systems and Technology (MSST), 2024
- **A Feature-Driven Fixed-Ratio Lossy Compression Framework for Real-World Scientific Datasets**
  Md Hasanur Rahman, Sheng Di, Kai Zhao, Robert Underwood, Guanpeng Li, Franck Cappello
  IEEE International Conference on Data Engineering (ICDE), 2023
  Acceptance rate: 19.1%
- **Peppa-X: Finding Program Test Inputs to Bound Silent Data Corruption Vulnerability in HPC Applications**
  Md Hasanur Rahman, Aabid Shamji, Shengjian Guo, Guanpeng Li
  ACM International Conference for High-Performance Computing, Networking, Storage and Analysis (SC), 2021
  Acceptance rate: 23.6%

Journal and Survey Publications

- **A Survey on Error-Bounded Lossy Compression for Scientific Datasets**
  Sheng Di, Jinyang Liu, Kai Zhao, Xin Liang, Robert Underwood, Zhaorui Zhang, Milan Shah, Yafan Huang, Jiajun Huang, Xiaodong Yu, Congrong Ren, Hanqi Guo, Grant Wilkins, Dingwen Tao, Jiannan Tian, Sian Jin, Zizhe Jian, Daoce Wang, Md Hasanur Rahman, Boyuan Zhang, Shihui Song, Jon Calhoun, Guanpeng Li, Kazutomo Yoshii, Khalid Alharthi, Franck Cappello
  ACM Computing Surveys, 2025
- **Investigating the Impact of Transient Hardware Faults on Deep Learning Neural Network Inference**
  Md Hasanur Rahman, Sabuj Laskar, Guanpeng Li
  Journal of Software Testing, Verification and Reliability (STVR), 2024

Co-Authored Peer-Reviewed Conference Publications

- **Significantly Improving Fixed-Ratio Compression Framework for Resource-Limited Applications**
  Tri Nguyen, Md Hasanur Rahman, Sheng Di, Michela Becchi
  International Conference on Parallel Processing (ICPP), 2024
  Acceptance rate: 29.0%
- **Characterizing Deep Learning Neural Network Failures between Algorithmic Inaccuracy and Transient Hardware Faults**
  Sabuj Laskar, Md Hasanur Rahman, Bohan Zhang, Guanpeng Li
  IEEE Pacific Rim International Symposium on Dependable Computing (PRDC), 2022
  Acceptance rate: 36.0%

Workshop Publications

- **Characterizing Spatial Data Traits for Modeling Generic Lossy Rate-Distortion Quality**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Parallel & Distributed Processing Symposium Workshops (IPDPS-W), 2025
- **LibPressio-Predict: Flexible and Fast Infrastructure for Inferring Compression Performance**
  Robert R. Underwood, Sheng Di, Sian Jin, Md Hasanur Rahman, Arham Khan, Franck Cappello

ACM International Workshop on Data Reduction for Big Scientific Data (DRBSD), in conjunction with SC, 2023

- **TensorFI+: A Scalable Fault Injection Framework for Modern Deep Learning Neural Networks**
  Sabuj Laskar, Md Hasanur Rahman, Guanpeng Li
  IEEE International Workshop on Resiliency, Security, Defences and Attacks (ISSRE-W), 2022

INDUSTRY EXPERIENCE

Samsung Research *2019–2021*
Software Engineer *Bangladesh*

· Developed production software components and application-level APIs in an industry research and development environment.
· Collaborated with engineering teams on software design, implementation, debugging, and testing for deployed systems.

HONORS AND AWARDS

- Graduate Research Excellence Award Finalist, 2025.
- SC Travel Grant, 2024.
- ICDE Travel Grant, 2023.

PROFESSIONAL SERVICE

- **Technical Program Committee:** IEEE CLOUD Summit, 2026.
- **Artifact Evaluation Committee:** IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2024.
- **Subreviewer:** ICS 2026; DSN 2026; HiPC 2025; ISSRE 2024, 2023, 2022; HPDC 2023, 2022; DSN 2023, 2022; Middleware 2022; SELSE 2022; PRDC 2021.
- **Student Volunteer:** SC 2024; ICDE 2023.

STUDENT MENTORING

- Ahmer Jamil, 2025–2026.
- Abdullah Naveed, 2024–2026.
- Sabuj Laskar, 2022–2023.